# VRTSTATISTICS

- Not limited to VR2Gather!

# GOALS

- Measurements of multi-user multi-machine XR experiences
- Very low complexity, easy to implement
- Low runtime impact
- Rich end-to-end results, down to individual events when needed

# RECORDING: FORMAT

```
stats: seq=23, ts=57386.256, component=OrchestratorController, orchestrator_ntptime_ms=1783605386227, …
stats: seq=114, ts=57397.075, component=VoicePipelineOther#1, …, preparer=AsyncVoicePreparer#0, …
stats: seq=237, ts=57398.367, component=AsyncVoicePreparer#0, fps=62.69, fps_dropped=0.00, …
```

- Lines of the form shown above, output by application under test
  - Simple to implement: print statement
  - No dependency on anything
- Can be recorded locally, on machine-under-test
- Records can be extracted from a general log
  - Because of stats: marker and key=value, fixed format

# RECORDING: CONTENTS

```
stats: seq=23, ts=57386.256, component=OrchestratorController, orchestrator_ntptime_ms=1783605386227, …
stats: seq=114, ts=57397.075, component=VoicePipelineOther#1, …, preparer=AsyncVoicePreparer#0, …
stats: seq=237, ts=57398.367, component=AsyncVoicePreparer#0, fps=62.69, fps_dropped=0.00, …
```

- Records clock relationship to other clocks in the whole system (first line)
- Records relationship between components (second line)
- Records measurements (third line)
- Granularity set by the experimenter, per experiment
  - Aggregated (e.g. every 10 s): overall statistics, negligible runtime impact
  - Per event: every frame or voice packet, so you can see where each one spent its time

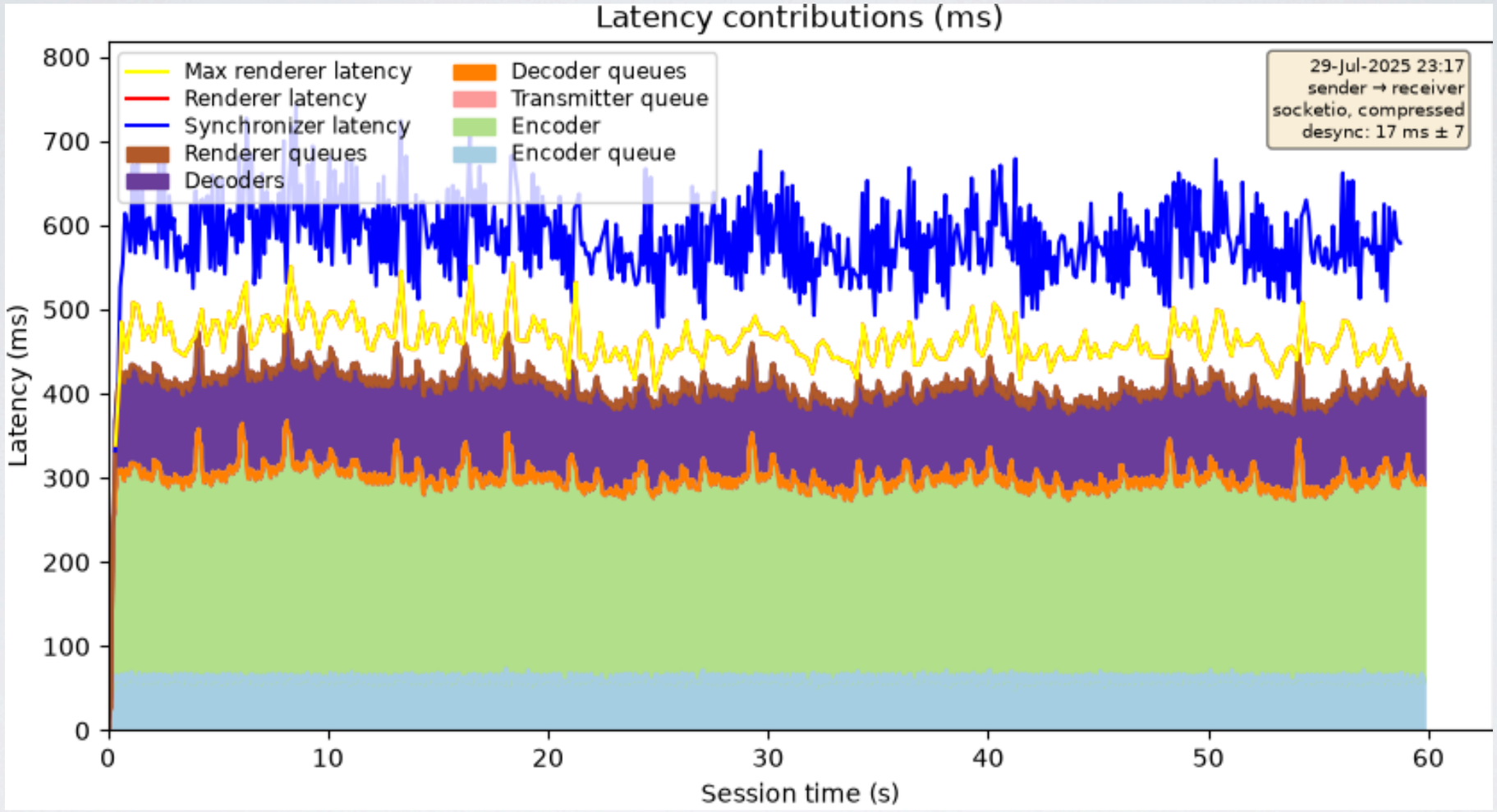
# DATA GATHERING

- Gathered from multiple sources afterwards
- Mapped to unified timeline
  - Using clock relationship records
- Combined into unified "database"
- Annotate records
  - Using component relationship records
  - Create roles p2.voice.preparer from implementation detail AsyncVoicePreparer#0

# ANALYSIS

- Select records (all voice going from p1 to p2)
- Project to a table (keep only columns latency and drops)
- Plot or analyse (Pandas, CSV export)

# EXAMPLE




- End-to-end latency for point clouds, from sender capture to receiver display
- Plus information on where that latency comes from
- And what was measured, with which uncertainty

# TRACK RECORD

- System experiments (latency, framerate, bandwidth) since the VR2Gather paper (Viola et al., IEEE MultiMedia 2023)

- User studies (gaze, speech, questionnaires), unpublished so far

- Used for a new transport by a partner: WebRTC SFU benchmarking with Ghent University (De Fré et al., NOSSDAV 2026)